

Experimental Characterization of the Modulation Port Bandwidth of a Signal Generator

Experimental Report

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Experiment: Modulation-Port Bandwidth Measurement

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1 Introduction

The purpose of this experiment is to characterize the frequency response and effective bandwidth of the external modulation port of a RIGOL DG4162 signal generator.

The modulation input is intended to be used as an actuator in a feedback control system. Therefore, it is important to determine how the modulation response changes as a function of modulation frequency.

In this experiment, the response is characterized by measuring:

1. the output amplitude as a function of modulation frequency,
2. the phase difference between the applied modulation signal and the resulting frequency modulation,

The experiment also uses a Red Pitaya STEMLab 125-14 FPGA to process the signals and convert the measured frequency variation into a voltage that can be directly observed using an oscilloscope.

2 Objective

The main objectives of the experiment are:

- To experimentally determine the frequency response of the modulation input of the RIGOL DG4162 signal generator.
- To measure the variation of output amplitude with modulation frequency.
- To measure the phase shift introduced by the modulation path as a function of frequency.
- To estimate the usable bandwidth of the modulation input.
- To implement a real-time frequency-to-voltage measurement system using the Red Pitaya STEMLab 125-14 FPGA.

3 Experimental Setup

The complete experimental arrangement used for the bandwidth measurement is shown in Fig. 1.

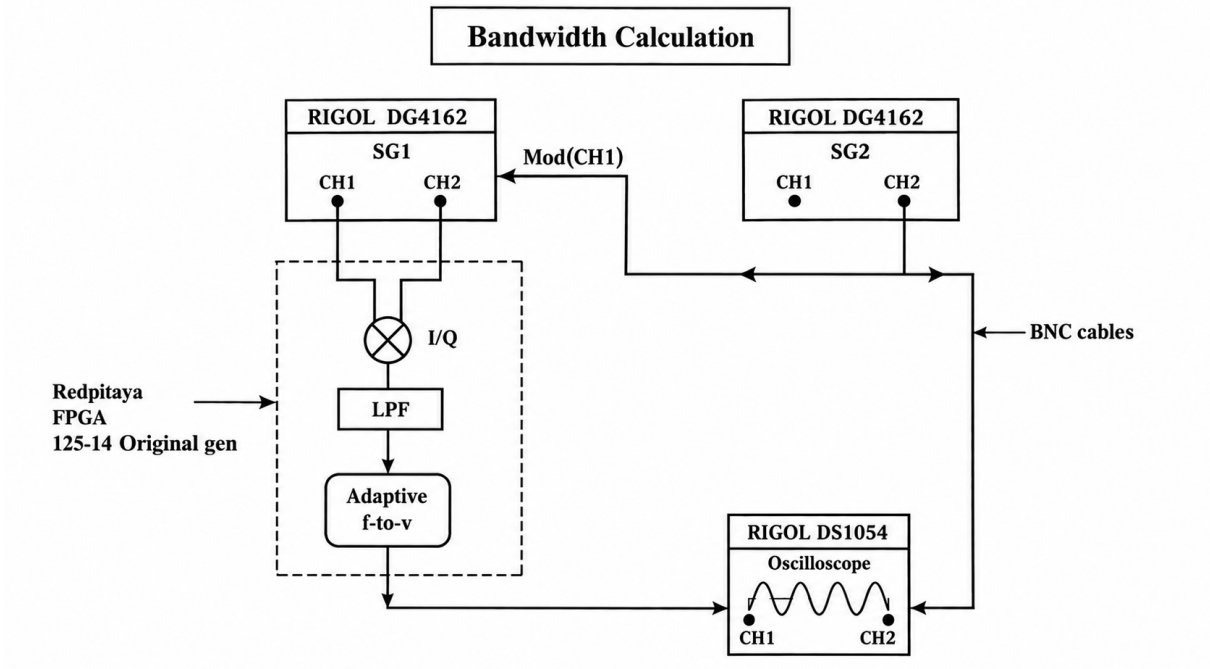


Figure 1: Experimental setup used for measurement of the modulation-port bandwidth.

The setup consists primarily of two RIGOL DG4162 signal generators, a Red Pitaya STEMLab 125-14 FPGA board, and a digital oscilloscope.

3.1 Signal Generator SG1

Signal generator SG1 is used as the controlled signal source.

Channel 1 and Channel 2 of SG1 are connected to the Red Pitaya for signal comparison and frequency measurement.

The external modulation input of SG1 is driven using the modulation signal generated by SG2. Here SG1's CH1 and CH2 settings are exactly same- Both are Sine wave, Frequency 20MHz, Amplitude 500mVpp. And Modulation settings for CH1 is- Modulation type-FM(Frequency Modulation), Source-External, FM Dev 500kHz

3.2 Signal Generator SG2

Signal generator SG2's CH2 is used to generate the modulation waveform. Here settings are Frequency- Varying from 10Hz to 400kHz, Amplitude=2.3Vpp

The modulation frequency is varied systematically during the experiment while keeping the remaining experimental parameters fixed.

One branch of the modulation signal is connected to the modulation input of CH1 Modulation port in SG1, while another branch is connected directly to the oscilloscope. This provides a reference signal for phase comparison.

3.3 Oscilloscope

A RIGOL DS1054 oscilloscope is used to simultaneously monitor the reference modulation signal and the output of the Red Pitaya.

Channel 1 of the oscilloscope is connected to:

OUT2 of the Redpitaya

Channel 2 of the oscilloscope is connected to:

Modulation waveform from SG2 CH2

The oscilloscope measurements are used to determine both the peak-to-peak voltage and the relative phase difference between the signals.

4 Red Pitaya FPGA Signal Processing

The Red Pitaya STEMLab 125-14 is used as a real-time digital signal processing platform.

The FPGA receives the two signals generated by SG1 and processes them using a sequence of digital signal-processing blocks.

The overall FPGA processing chain is

$$\text{ADC Inputs} \rightarrow \text{I/Q} \rightarrow \text{Low-Pass Filter} \rightarrow \text{Adaptive F-to-V Converter} \rightarrow \text{DAC Output.} \quad (1)$$

4.1 Red Pitaya Hardware

The FPGA implementation is performed on a Red Pitaya STEMLab 125-14 Original Generation board.

Important hardware parameters are:

Table 1: Red Pitaya hardware parameters.

Parameter	Value
FPGA device	Xilinx Zynq-7010
ADC resolution	14 bit
ADC sampling frequency	125 MHz
DAC resolution	14 bit
DAC sampling frequency	125 MHz
Input channels	2
Output channels	2

4.2 I/Q Signal Mixing

Two sinusoidal signals of approximately 20 MHz, each with an amplitude of 500 mVpp, are applied to the two input channels of the Red Pitaya.

The two input signals may be written as

$$V_1(t) = A \sin(2\pi f_1 t + \phi_1) \quad (2)$$

and

$$V_2(t) = A \sin(2\pi f_2 t + \phi_2), \quad (3)$$

where A corresponds to the 500 mVpp input amplitude.

The purpose of the I/Q stage is to compare these two high-frequency signals and extract their relative frequency and phase information.

One of the signals is taken as the reference. From this reference, two signals are formed that are separated by 90° in phase:

$$\sin(2\pi f_2 t + \phi_2) \quad (4)$$

and

$$\cos(2\pi f_2 t + \phi_2). \quad (5)$$

These two orthogonal reference signals are used for two separate digital multiplications. For the in-phase channel,

$$I_{\text{mix}}(t) = V_1(t) \sin(2\pi f_2 t + \phi_2). \quad (6)$$

Using the identity

$$\sin A \sin B = \frac{1}{2} [\cos(A - B) - \cos(A + B)], \quad (7)$$

the result contains a difference-frequency term and a sum-frequency term. After low-pass filtering, the high-frequency sum term is removed, giving

$$I(t) \propto \cos [2\pi(f_1 - f_2)t + (\phi_1 - \phi_2)]. \quad (8)$$

For the quadrature channel,

$$Q_{\text{mix}}(t) = V_1(t) \cos(2\pi f_2 t + \phi_2). \quad (9)$$

After multiplication and low-pass filtering, the quadrature component is approximately

$$Q(t) \propto \sin [2\pi(f_1 - f_2)t + (\phi_1 - \phi_2)]. \quad (10)$$

Thus, the I and Q signals represent the same beat signal but are separated from each other by approximately 90° .

If the two input frequencies are not exactly equal, the I and Q values continuously rotate in the I-Q plane at the beat frequency

$$f_{\text{beat}} = |f_1 - f_2|. \quad (11)$$

If the two input frequencies are equal, the beat frequency becomes zero and the I and Q values become nearly constant. Their values then contain the relative phase information between the two input signals.

The relative phase can be obtained from

$$\Delta\phi = \text{atan2}(Q, I). \quad (12)$$

Therefore, the I/Q stage performs two important functions: it down-converts the two signals near 20 MHz to their much lower difference frequency, and it preserves the relative phase information between them. This makes the resulting signal suitable for further processing by the digital low-pass filter and the adaptive frequency-to-voltage converter.

4.3 Digital Low-Pass Filter

After the I/Q mixing stage, the mixer outputs contain both the difference-frequency component and the sum-frequency sum component.

For two input signals near 20 MHz, the mixing process produces

$$f_{\text{difference}} = |f_1 - f_2| \quad (13)$$

and

$$f_{\text{sum}} = f_1 + f_2 \approx 40 \text{ MHz}. \quad (14)$$

The difference-frequency component contains the useful beat-frequency information, whereas the sum-frequency component is unwanted. Therefore, a digital low-pass filter is applied after the I/Q mixing stage.

In the present FPGA design, the low-pass filter has a bandwidth of approximately

$$\boxed{f_{\text{LPF}} \approx 2.04 \text{ MHz}} \quad (15)$$

The filter therefore allows beat-frequency components from low frequency up to approximately 2.04 MHz to pass, while strongly attenuating the high-frequency mixing component near 40 MHz.

The filtered I and Q signals can be written approximately as

$$I(t) \propto \cos [2\pi(f_1 - f_2)t + \Delta\phi] \quad (16)$$

and

$$Q(t) \propto \sin [2\pi(f_1 - f_2)t + \Delta\phi], \quad (17)$$

where

$$\Delta\phi = \phi_1 - \phi_2. \quad (18)$$

Thus, the purpose of the low-pass filter is to remove the unwanted high-frequency sum-frequency term and retain the lower-frequency I/Q beat signal containing the relative frequency and phase information of the two input signals.

Since the filter bandwidth is approximately 2.04 MHz, the I/Q processing path can preserve beat-frequency variations over a much wider range than a narrow-band filter, which is important for following the rapidly changing frequency difference used in this experiment.

4.4 Adaptive Frequency-to-Voltage Converter

After the I/Q stage and low-pass filtering, the FPGA estimates the signed frequency difference between the two input signals,

$$\Delta f = f_{\text{IN2}} - f_{\text{IN1}}. \quad (19)$$

The purpose of the adaptive frequency-to-voltage converter is to convert this frequency difference into a proportional voltage at Red Pitaya OUT2.

The selected full-scale mapping is

$$-2 \text{ MHz} \rightarrow -1 \text{ V}, \quad (20)$$

$$0 \text{ Hz} \rightarrow 0 \text{ V}, \quad (21)$$

and

$$+2 \text{ MHz} \rightarrow +1 \text{ V}. \quad (22)$$

Therefore, the frequency-to-voltage sensitivity is

$$K_{FV} = \frac{1 \text{ V}}{2 \times 10^6 \text{ Hz}} = \boxed{0.5 \text{ } \mu\text{V/Hz}}. \quad (23)$$

The ideal output-voltage relation is therefore

$$\boxed{V_{\text{OUT2}} = K_{FV} \Delta f = \Delta f \times 0.5 \text{ } \mu\text{V/Hz}} \quad (24)$$

For example,

$$\Delta f = +500 \text{ kHz} \quad (25)$$

produces approximately

$$V_{\text{OUT2}} = +0.25 \text{ V}, \quad (26)$$

while

$$\Delta f = -500 \text{ kHz} \quad (27)$$

produces approximately

$$V_{\text{OUT2}} = -0.25 \text{ V}. \quad (28)$$

Thus, the sign of the output voltage directly indicates which input frequency is higher.
If

$$f_{\text{IN2}} > f_{\text{IN1}}, \quad (29)$$

OUT2 becomes positive, whereas if

$$f_{\text{IN2}} < f_{\text{IN1}}, \quad (30)$$

OUT2 becomes negative. When both frequencies are equal,

$$\Delta f \approx 0, \quad (31)$$

and therefore

$$V_{\text{OUT2}} \approx 0. \quad (32)$$

The Red Pitaya DAC uses a signed 14-bit digital value. The corresponding digital scaling is approximately

$$\boxed{\text{DAC}_{\text{code}} = \Delta f_{\text{Hz}} \frac{8192}{2\,000\,000}} \quad (33)$$

or

$$\text{DAC}_{\text{code}} \approx 0.004096 \Delta f_{\text{Hz}}. \quad (34)$$

For example, for a frequency difference of 500 kHz,

$$\text{DAC}_{\text{code}} = 500000 \times 0.004096 = 2048, \quad (35)$$

which corresponds to approximately 0.25 V at OUT2.

4.4.1 Adaptive Averaging

The frequency estimation is made adaptive in order to obtain both fast response and low noise.

The estimator can use

$$\boxed{M = 1, 2, 4, 8, 16} \quad (36)$$

samples for averaging.

When the measured frequency difference is changing rapidly, a smaller value of M is selected. This reduces the averaging time and allows the converter to respond quickly to fast frequency changes.

When the frequency difference becomes smaller or more stable, a larger value of M is used. This increases the averaging and reduces noise, producing a cleaner and more stable output voltage.

Hysteresis is included in the switching logic so that the estimator does not continuously switch between adjacent averaging settings.

Therefore, the complete signal-processing path is

$$\boxed{\text{IN1, IN2} \rightarrow I/Q \rightarrow \Delta\phi \rightarrow \Delta f \rightarrow \text{Adaptive Averaging} \rightarrow \text{F-to-V Scaling} \rightarrow \text{OUT2}} \quad (37)$$

while OUT1 is kept at zero.

The main advantage of this adaptive converter is that it provides a wide signed measurement range of approximately ± 2 MHz while maintaining faster response for rapidly changing frequency differences and improved noise performance when the frequency difference is more stable.

4.5 FPGA Block Diagram

The complete signal-processing architecture implemented inside the Red Pitaya FPGA is shown in Fig. 2.

Internal FPGA Signal-Processing Architecture

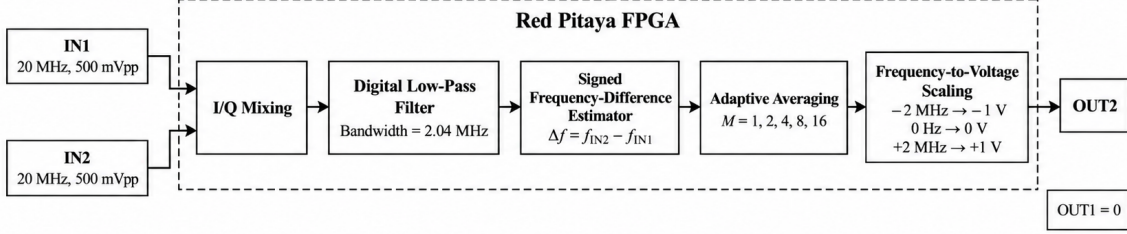


Figure 2: Internal FPGA signal-processing architecture implemented on the Red Pitaya STEMLab 125-14.

The two input signals are first processed by the I/Q stage. The resulting components are passed through a digital low-pass filter with a bandwidth of approximately 2.04 MHz. The filtered I/Q signals are then used to estimate the signed frequency difference between the two input signals. Adaptive averaging is subsequently applied using $M = 1, 2, 4, 8, 16$, depending on the rate of change of the measured frequency difference. Finally, the estimated frequency difference is converted into a proportional voltage with a scaling of

$$-2 \text{ MHz} \rightarrow -1 \text{ V}, \quad +2 \text{ MHz} \rightarrow +1 \text{ V}, \quad (38)$$

and the resulting signal is sent to OUT2. OUT1 is maintained at zero.

5 Experimental Procedure

To determine the modulation-port bandwidth, a sinusoidal modulation signal is applied to the external modulation input of SG1's CH1.

The modulation frequency is varied from 10Hz to 400kHz.

For every modulation frequency, two quantities are measured:

1. Output amplitude V_{pp} ,
2. Phase difference ϕ between the reference modulation signal and the measured response, To measure nicely modulation signal is triggered.

Thus, the modulation system can be characterized by the transfer function

$$H(f) = \frac{V_{out}(f)}{V_{in}(f)}. \quad (39)$$

6 Amplitude Response

The first measurement investigates the variation of the output peak-to-peak voltage with modulation frequency.

6.1 Measured Data

The measured output voltage as a function of modulation frequency is listed in Table 2. The modulation frequency was varied from 0.01 kHz to 450 kHz, while the corresponding output peak-to-peak voltage was measured using the oscilloscope.

Table 2: Measured amplitude response of the modulation port.

Frequency (kHz)	Output V_{pp} (mV)
0.01	232
0.02	232
0.05	234
0.10	228
0.20	228
0.50	228
1	228
5	228
8	228
10	228
20	226
30	224
40	222
50	222
100	218
130	216
150	212
180	204
190	204
194	202
210	194
220	192
250	182
260	176
300	160
320	142
330	137
350	130
380	120
400	100
450	86

6.2 Frequency vs Amplitude

Figure 3 shows the measured peak-to-peak voltage as a function of modulation frequency. At low modulation frequencies, the output voltage remains approximately constant at 228 mV–234 mV. As the modulation frequency is increased, the output amplitude gradually decreases, indicating the finite frequency response of the modulation port.

A more significant reduction in amplitude is observed above approximately 200 kHz. At 300 kHz, the measured voltage is approximately 160 mV, while at 450 kHz it decreases to approximately 86 mV.

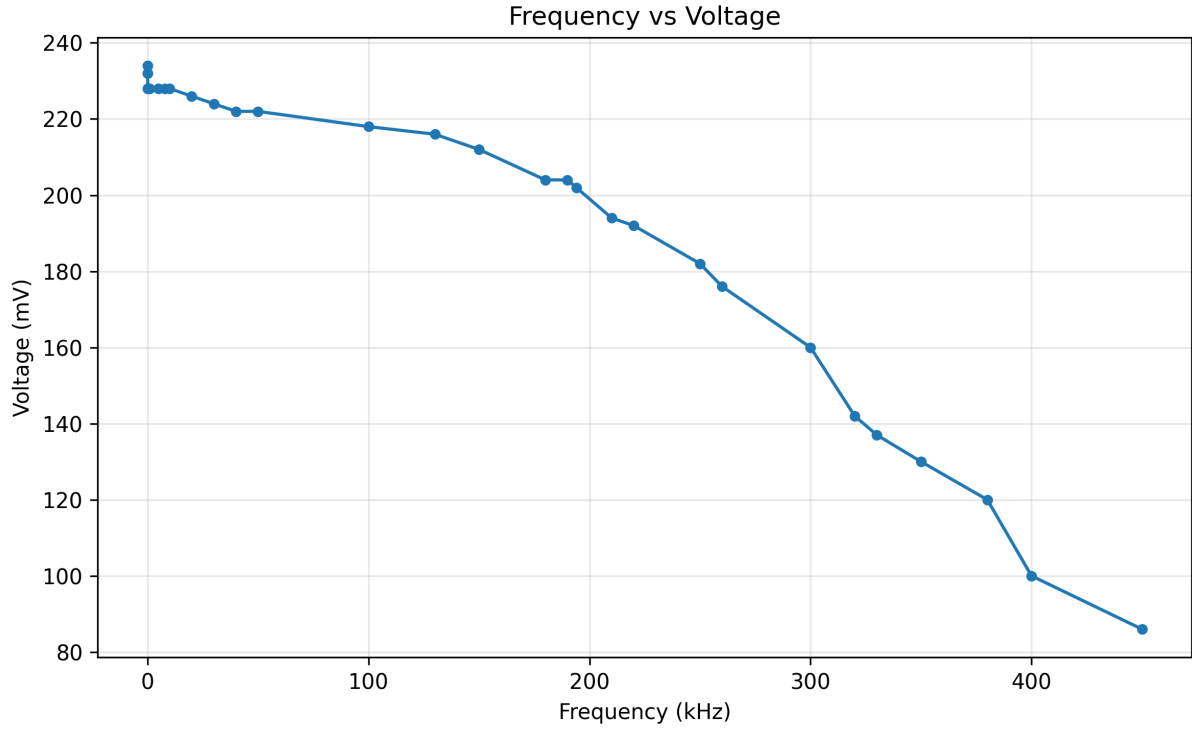


Figure 3: Measured peak-to-peak output voltage as a function of modulation frequency.

7 Phase Response

The phase difference between the reference modulation signal and the measured response was recorded as the modulation frequency was varied.

7.1 Measured Phase Data

The measured phase response is listed in Table 3.

Table 3: Measured phase response of the modulation port.

Frequency (kHz)	Phase difference (degree)
0.01	179.3
0.02	180.0
0.05	179.1
0.10	179.3
0.20	179.3
0.50	179.1
1	179.3
5	175.5
8	173.1
10	172.1
20	162.0
30	152.0
40	142.5
50	135.0
100	90.0
130	60.0
150	40.0
180	12.0
190	2.0
194	1.3
210	21.0
220	24.3
250	38.3
260	145.0
300	112.0
320	165.0
330	140.0
350	74.0

7.2 Frequency vs Phase

Figure 4 shows the measured phase difference as a function of modulation frequency.

At very low modulation frequencies, the measured phase remains close to 180° . As the modulation frequency is increased, the phase difference decreases gradually. For example, the measured phase is approximately 172.1° at 10 kHz, 135° at 50 kHz, and 90° at 100 kHz.

The phase continues to decrease and reaches approximately 1.3° at 194 kHz. Above this frequency, the measured phase appears to increase again. This behaviour is associated with the periodic nature of phase measurement: phase is normally displayed within a limited angular interval, so once the relative phase passes through a complete phase cycle, the measured value can wrap around.

Therefore, the apparent discontinuities at higher frequencies should not be interpreted as an abrupt physical change in the modulation response. Instead, the phase data should be considered with phase wrapping taken into account.

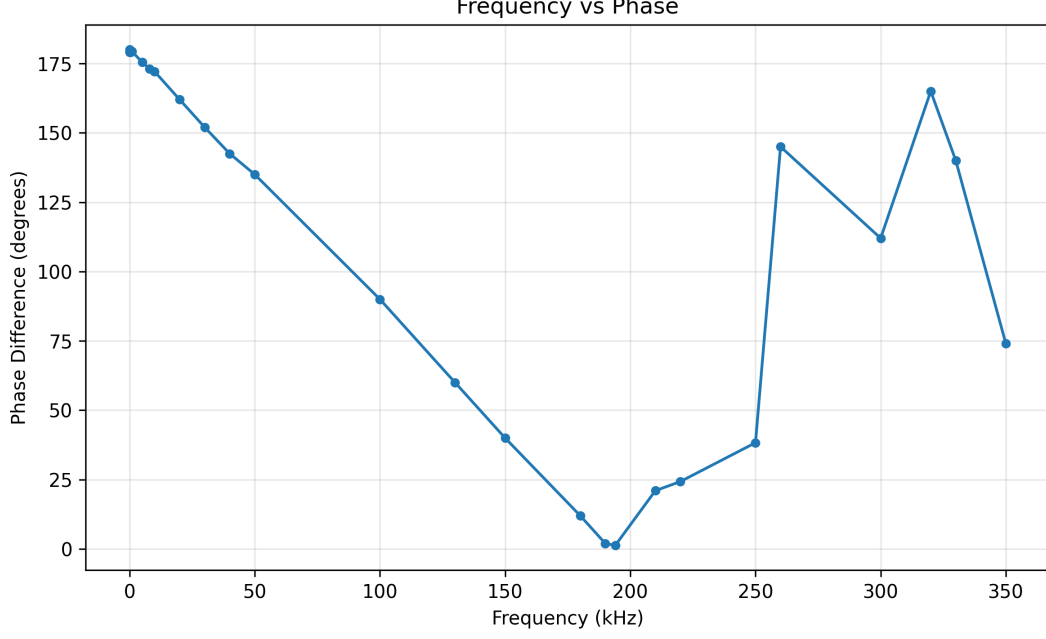


Figure 4: Measured phase difference as a function of modulation frequency.

8 Determination of Modulation Bandwidth

The bandwidth of the modulation port is determined from the measured amplitude response. In the low-frequency flat-response region, the measured output voltage is approximately

$$V_{pp}(f_{ref}) \approx 228 \text{ mV}. \quad (40)$$

The normalized amplitude response is defined as

$$A_{norm}(f) = \frac{V_{pp}(f)}{V_{pp}(f_{ref})}. \quad (41)$$

The corresponding gain in decibels is

$$G(f) = 20 \log_{10} \left[\frac{V_{pp}(f)}{V_{pp}(f_{ref})} \right]. \quad (42)$$

For the conventional -3 dB bandwidth, the output amplitude becomes

$$\frac{V_{pp}(f_{BW})}{V_{pp}(f_{ref})} = \frac{1}{\sqrt{2}} \approx 0.707. \quad (43)$$

Therefore, the expected voltage at the -3 dB point is

$$V_{pp}(f_{BW}) = 0.707 \times 228 \text{ mV}, \quad (44)$$

which gives

$$\boxed{V_{pp}(f_{BW}) \approx 161.2 \text{ mV}} \quad (45)$$

From the experimental data, the measured output voltage at 300 kHz is approximately 160 mV. The corresponding gain is

$$G(300 \text{ kHz}) = 20 \log_{10} \left(\frac{160}{228} \right), \quad (46)$$

giving

$$\boxed{G(300 \text{ kHz}) \approx -3.08 \text{ dB}} \quad (47)$$

which is very close to the conventional -3 dB cutoff condition.

Therefore, the experimentally measured modulation-port bandwidth is

$$\boxed{f_{\text{BW}} \approx 300 \text{ kHz}} \quad (48)$$

Thus, the modulation input maintains approximately its low-frequency response up to a few hundred kilohertz, with the -3 dB point occurring near 300 kHz.

9 Experimental Results

The main experimental results obtained from the amplitude and phase measurements are summarized in Table 4.

Table 4: Summary of experimental results.

Parameter	Measured value
Carrier frequency	20 MHz
Applied signal amplitude	500 mVpp
Frequency range investigated	0.01 kHz – 450 kHz
Low-frequency output amplitude	≈ 228 mVpp
-3 dB amplitude	≈ 161.2 mVpp
Measured modulation bandwidth	≈ 300 kHz
Measured phase at 300 kHz	$\approx 112^\circ$
Maximum frequency investigated	450 kHz

The low-frequency response remains approximately constant at 228 mVpp. The corresponding -3 dB amplitude is approximately 161.2 mVpp. At a modulation frequency of 300 kHz, the measured output amplitude is approximately 160 mVpp, giving a gain of approximately -3.08 dB. Therefore, the experimentally determined modulation bandwidth is approximately

$$\boxed{f_{\text{BW}} \approx 300 \text{ kHz}} \quad (49)$$

At this frequency, the measured phase difference is approximately

$$\boxed{\phi(300 \text{ kHz}) \approx 112^\circ.} \quad (50)$$

The amplitude and phase measurements together characterize the frequency-dependent response of the modulation path. The amplitude response is used to determine the conventional -3 dB bandwidth, while the phase response provides information about the phase delay introduced by the complete experimental system.

10 Discussion

The experimentally measured amplitude and phase responses provide information about the dynamic behaviour of the modulation input.

At low modulation frequencies,

As the modulation frequency increases,

The reduction in amplitude at higher frequencies indicates the finite bandwidth of the internal modulation electronics.

The measured phase lag also increases with frequency, indicating the presence of delay and filtering within the modulation path.

11 Sources of Experimental Uncertainty

Possible sources of uncertainty in the experiment include:

- Oscilloscope cursor resolution.
- Uncertainty in measurement of Δt .
- Variation of signal amplitude with frequency.
- BNC cable propagation delay.
- Signal-generator output characteristics.
- Sampling and quantization effects in the Red Pitaya ADC.
- Digital-filter phase delay inside the FPGA.
- Response time of the frequency-to-voltage converter.

The phase uncertainty resulting from an uncertainty δt in the measured time difference can approximately be written as

$$\delta\phi = 360^\circ f \delta t. \quad (51)$$

12 Relevance to the Feedback Control System

The modulation port is intended to act as the frequency actuator in a closed-loop frequency/phase control system.

In such a system, the controller output voltage is applied to the modulation input of the signal generator.

The relationship can be represented as

$$\Delta f = K_{\text{FM}} V_{\text{control}}, \quad (52)$$

where K_{FM} is the frequency tuning coefficient of the signal generator.

The finite modulation bandwidth therefore places an upper limit on the frequency range over which the actuator can efficiently respond to corrections produced by the feedback controller.

13 Conclusion

In this experiment, the frequency response of the external modulation input of a RIGOL DG4162 signal generator was experimentally characterized.

The response was measured in terms of both amplitude and phase as the modulation frequency was varied.

The Red Pitaya STEMLab 125-14 FPGA was used to implement the signal-processing chain required to extract the frequency information and convert it into an observable voltage.

The measured frequency response indicates that

The experimentally determined modulation bandwidth is

$$\boxed{f_{\text{BW}} = [300\text{kHz}]} \quad (53)$$

This measurement is important for determining the maximum practical feedback bandwidth achievable when the modulation input is used as an actuator in the planned phase/frequency locking system.

A FPGA Implementation Details

This appendix contains the detailed implementation of the Red Pitaya FPGA processing chain.

A.1 FPGA Clock

The FPGA processing operates using a clock frequency of

$$f_{\text{clk}} = 125 \text{ MHz}. \quad (54)$$

The corresponding FPGA clock period is

$$T_{\text{clk}} = \frac{1}{f_{\text{clk}}} = 8 \text{ ns}. \quad (55)$$

A.2 I/Q Implementation

A.3 Digital Low-Pass Filter

A.4 Adaptive Frequency-to-Voltage Algorithm

A.5 DAC Scaling

For a 14-bit DAC, the digital FPGA result is converted into the appropriate DAC output code before being sent to the physical output.

B Additional Experimental Data

References

- [1] RIGOL Technologies, *DG4000 Series Function/Arbitrary Waveform Generator User's Guide*.
- [2] Red Pitaya, *STEMlab 125-14 Documentation*.